

TANSHENG ZHU

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EDUCATION

Tsinghua University <i>Ph.D. Student in Computer Science, Institute for Interdisciplinary Information Sciences</i> Advisor: Prof. Jian Li • Research Interests: <i>Machine Learning Theory, Large Language Models, Optimization</i>	Aug. 2025 – Present Beijing, China
Shanghai Jiao Tong University <i>B.S. (Honors) in Mathematics & Minor in Finance, Zhiyuan College</i> Advisor: Prof. Zhenli Xu • Shanghai Outstanding Graduate (Top 1%); Best Bachelor Thesis of SJTU (Top 1%)	Aug. 2021 – June 2025 Shanghai, China
The Chinese University of Hong Kong <i>Exchange Student Junior Research Assistant, Department of Mathematics</i> Advisor: Prof. Fenglei Fan	Jan. 2024 – Sept. 2024 Hong Kong, China

RESEARCH EXPERIENCE

Scaling Limits of Infinitely Deep Neural Networks Advisor: Prof. Fenglei Fan, The Chinese University of Hong Kong • Established the theoretical framework of infinitely deep neural networks by weakly dependent random processes. • Leveraged the ergodic theory to interpret the width-depth equivalence of neural networks.	Jan. 2024 – Aug. 2025
Density-based Exploration Strategy for Scalable Bayesian Optimization Advisor: Prof. Zhenli Xu, Shanghai Jiao Tong University • Proposed the BOKE algorithm with guarantees, generalized Bayesian optimization into non-GP modeling. • Conducted experiments to verify its competitive performance in space-filling design and black-box optimization.	Apr. 2024 – June 2025
Monte-Carlo Tree Search in Continuous Action Spaces via Kernel Regression Advisor: Prof. Zhenli Xu, Shanghai Jiao Tong University • Proposed the BOKR-MCTS algorithm to leverage Monte-Carlo tree search into continuous state-action spaces. • Conducted experiments to show its computational efficiency and competitive performance in online planning tasks.	Nov. 2022 – Apr. 2024

PUBLICATIONS

- [1] S. Wang*, Y. Zheng*, **T. Zhu***, S. Chen*, S. Liu, S. Zheng, and L. Jian, “AdaRoPE: Not all attention heads should rotate and scale equally,” in *Proceedings of the International Conference on Machine Learning*, 2026.
- [2] S. Wang*, Y. Dong*, R. Chang*, **T. Zhu***, Y. Sun, K. Lyu, and J. Li, “When bias pretends to be truth: How spurious correlations undermine hallucination detection in LLMs,” 2025. arXiv: 2511.07318 [cs.CL], Preprint.
- [3] **T. Zhu**, H. Zhou, K. Jin, X. Xu, Q. Yuan, and L. Ji, “Bayesian optimization by kernel regression and density-based exploration,” 2025. arXiv: 2502.06178 [math.OG], Preprint.
- [4] K. Jin, **T. Zhu**, H. Zhou, X. Xu, Q. Yuan, and L. Ji, “Bayesian-optimized progressive widening via kernel regression for Monte Carlo tree search in continuous action spaces,” in *23rd IFAC Symposium on Automatic Control in Aerospace (ACA 2025)*, vol. 59, Elsevier, 2025, pp. 1291–1296.

SELECTED AWARDS

Winning Prize in S.-T. Yau College Student Mathematics Contest, Appl. & Comp. Mathematics Track	May 2024
Finalist Winner in Mathematical Contest in Modeling (Top 3% of 18000+ teams)	May 2024
1st Prize in Shanghai Region of Chinese Mathematics Competitions	Nov. 2023
1st Prize in Huashu Cup National Mathematical Contest in Modeling (solo winner)	Aug. 2023
1st Prize in Chinese High School Mathematics League	Oct. 2020

SKILLS

Programming Languages: Python, Matlab, C/C++	
Leadership: Leadership Team Member, <i>Mathematical Modelling Association of SJTU</i>	2022 – 2024
Class President, <i>Zhiyuan College, Shanghai Jiao Tong University</i>	2021 – 2023